5.18 Construction of indexing head

The construction of a universal dividing head as shown in figure and explained below.

Base

The base of the indexing head is fitted in the 'T' – slots of the milling machine table. It supports all the other parts of dividing head.

Spindle

The spindle is situated at the centre of the dividing head. It has a taper hole to receive a live center. The spindle is supported on a swiveling block, which makes the spindle to be tilted through any angle from 5^0 below horizontal to 10^0 beyond vertical. A worm wheel is mounted on the spindle. While doing direct or rapid indexing the index plate is directly fitted on the front end of the spindle nose.

Worm shaft

It is situated at right angles to the main spindle of the dividing head. A single threaded worm is mounted on the worm shaft which meshes with the worm wheel.

An indexing plate is fitted on the front end of the worm shaft and with the help of a handle, the worm shaft can be rotated to a pre determined amount.

Indexing Plate

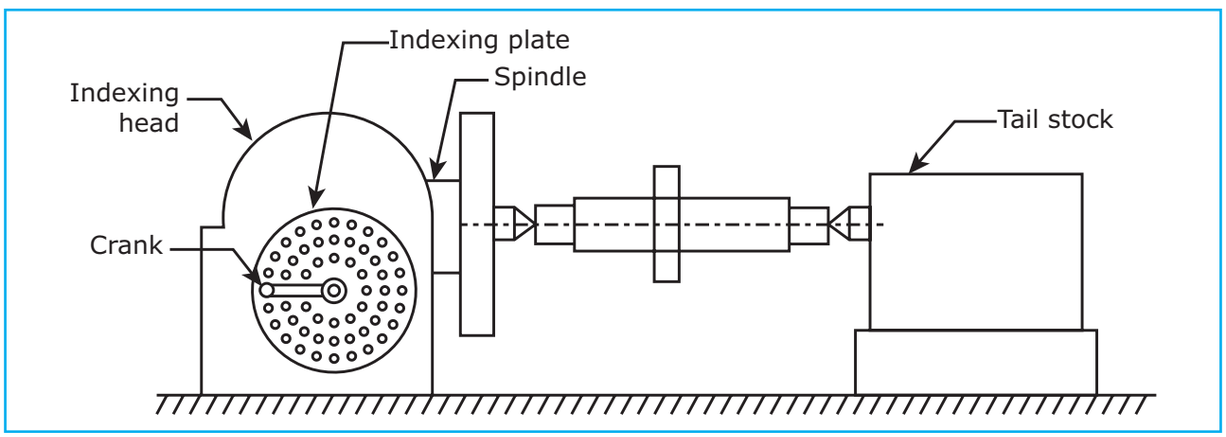
It is mounted on the front end of the worm shaft. It is circular disk having different numbers of equally spaced holes arranged in concentric circles. The crank is positioned in the required hole circle and rotated through a calculated amount while indexing. The sector arm is used to eliminate the necessity of counting the holes on the index plate each time the index crank is moved.

Tail Stock

The work is held between the center of the spindle and the tailstock. It can be made slide and positioned at the required location.

Working principle of dividing head

When the crank is rotated with help of a handle through the required number of holes in the index plate. The work is rotated to required amount. This is possible because of the worm and worm wheel mechanism.



Construction of an indexing head

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